Derivation of the deformed generating function $H^{(c)}(t) = \dfrac{e^{ct}}{1-t^2}$

Question

Take the sequence with $h_0 = 1$ and $h_r = 2$ for $r \geq 1$. Its generating function is $H(t) = \frac{1+t}{1-t}$, which forces the paired sequence $j = (2, 0, 2, 0, \dots)$. The c-deformation prepends c to j. Working through equation (D), show that the deformed generating function is

$$H^{(c)}(t) = \frac{e^{ct}}{1-t^2}.$$

The relation (D)

Everything rests on equation (D) written as a logarithmic derivative. With $H(t) = \sum_{n\geq 0} h_n t^n$ and $J(t) = \sum_{r\geq 1} j_r t^r$, the following three statements are equivalent:

$$J(t) = t\,\frac{d}{dt}\log H(t), \qquad tH'(t) = H(t)\,J(t), \qquad nh_n = \sum_{r=1}^{n} j_r\,h_{n-r} \quad (n \geq 1).$$

The middle identity follows by matching the coefficient of t^n on each side of the first; the third is that coefficient identity written out.

Step 1: the base J

With $h_0 = 1$ and $h_r = 2$ for $r \geq 1$,

$$H(t) = 1 + 2\sum_{r\geq 1} t^r = 1 + \frac{2t}{1-t} = \frac{1+t}{1-t}.$$

Hence

$$\frac{d}{dt}\log H(t) = \frac{d}{dt}\big[\log(1+t) - \log(1-t)\big] = \frac{1}{1+t} + \frac{1}{1-t} = \frac{2}{1-t^2},$$

and therefore

$$J(t) = t\cdot\frac{2}{1-t^2} = \frac{2t}{1-t^2} = 2t + 2t^3 + 2t^5 + \cdots,$$

so that $j = (2, 0, 2, 0, \dots)$, i.e. $j_r = 2$ for odd r and 0 for even r.

Step 2: prepending c to a generating function

The deformation sends a sequence $S = \{x_1, x_2, \dots\}$ to $S^{(c)} = \{c, x_1, x_2, \dots\}$: index 1 becomes c, and each old entry x_r moves to index $r+1$. If $X(t) = \sum_{r\geq 1} x_r t^r$, then

$$X^{(c)}(t) = c\,t + \sum_{r\geq 2} x_{r-1} t^r = c\,t + t\sum_{s\geq 1} x_s t^s = c\,t + t\,X(t).$$

Applying this to j,

$$J^{(c)}(t) = c\,t + t\,J(t) = c\,t + \frac{2t^2}{1-t^2}.$$

Step 3: recover $H^{(c)}$ from $J^{(c)}$

The deformed sequence $h^{(c)}$ is the one paired with $j^{(c)}$ through the same relation (D), so

$$\frac{d}{dt}\log H^{(c)}(t) = \frac{J^{(c)}(t)}{t} = c + \frac{2t}{1-t^2}.$$

Integrating, and using $\displaystyle\int \frac{2t}{1-t^2}\,dt = -\log(1-t^2)$ (substitute $u = 1-t^2$, $du = -2t\,dt$),

$$\log H^{(c)}(t) = c\,t - \log(1-t^2) + C.$$

The constant C is fixed by the normalization $h_0^{(c)} = 1$, i.e. $H^{(c)}(0) = 1$, hence $\log H^{(c)}(0) = 0$. Evaluating the right-hand side at $t = 0$ gives $C = 0$. Exponentiating,

$$\boxed{H^{(c)}(t) = \frac{e^{ct}}{1-t^2}\,.}$$

Remark on the integration constant

The antiderivative $\int \frac{2t}{1-t^2}\,dt$ is determined only up to the branch choice $-\log(1-t^2)$ versus $-\log(t^2-1)$, which differ by a constant and produce $e^{ct}/(1-t^2)$ versus $e^{ct}/(t^2-1)$. These differ by an overall sign. The boundary condition $H^{(c)}(0) = 1$ selects the former: at $t = 0$ it equals $+1$, whereas $e^{ct}/(t^2-1)$ equals -1. A symbolic integrator that fixes its own constant of integration may return the second form; the physically correct branch is $e^{ct}/(1-t^2)$.

Consistency check

One verifies directly that $H^{(c)}(t) = e^{ct}/(1-t^2)$ reproduces $J^{(c)}(t)$ through relation (D):

$$t\,\frac{d}{dt}\log H^{(c)}(t) = t\,\frac{d}{dt}\big[c\,t - \log(1-t^2)\big] = t\left(c + \frac{2t}{1-t^2}\right) = c\,t + \frac{2t^2}{1-t^2} = J^{(c)}(t),$$

as required.